Sindhudurg Shikshan Prasarak Mandal's College of Engineering, Kankavli

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Computer Engineering, Computer Science & Engineering, Computer Science & Engineering (AIML)

Experiment No -7

Year: 2026-27	Class/Div.-TE	SEM: V
Course Code: 2115311	Course Name: Open Source Technology	
Branch: Computer Engineering, Computer Science & Engineering & AIML (Computer Science Engineering)		

Aim

System Administration
Modifyfile permissionprofiles (chmod, chown); Execute background software installation and environment setup using package managers (apt/yum).

Software Requirements

OS: Linux (apt or yum/dnfbased)
Privileges: root or sudoaccess
Utilities:chmod, chown,coreutils, backgroundjobcontrols(nohup,&,ps)
Network: OutboundHTTP/HTTPS accesstopackagerepositories

Theory

1. File Permissions & Ownership Mechanics

Linux enforces multi-user security through **Ownership** and **Access Modes.** Every file and directory is owned by a specific **User** (u) and **Group** (g), with permissions categorized for **Others** (o).

Permission Representation (Octal vs Symbolic)
Permissions consist of three capabilities: **Read (r=4), Write (w=2), and Execute (x=1).**

Category / Field	File Type	Owner (u)	Group (g)	Others (o)
Permissions	-	r w x	r - x	r - -
Binary / Decimal Calculation	—	4 + 2 + 1	4 + 0 + 1	4 + 0 + 0
Octal Value	—	**7**	**5**	**4**

chmod (Change Mode): Modifies permission bits. In symbolic mode, operations use operators (+, -, =) targeting entities (u, g, o, a). In octal mode, three digits set absolute access levels (4+2+1).

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	▯ **chown (Change Owner):** Alters user ownership (user) and optional group ownership (user:group) within the kernel's inode metadata structure. **2. Unattended & Background Package Execution** Standard package management runs interactively in the foreground. Unattended automation requires two mechanisms: 1. **Non-interactive Execution Flags (-y / DEBIAN_FRONTEND=noninteractive):** Prevents the process from hanging while waiting for user confirmation or TTY configuration prompts. 2. **Process Decoupling (nohup / & / Subshells):** o &: Sends the command process to run in the background within the current shell session. o nohup (No Hangup): Intercepts the SIGHUP (Signal Hangup) sent to processes when a terminal session closes, allowing jobs to complete cleanly in the background. o **I/O Redirection (> file 2>&1):** Redirects stdout and stderr streams to a log file or /dev/null so stdout does not block the background process.
Procedure	**1.Verify User Privileges and Network Connectivity:** Prerequisite Verification. Confirm elevated privileges and connectivity to package mirrors before launching processes. # Verify sudo privilege sudo -v # Verify repository connectivity ping -c 2 archive.ubuntu.com || ping -c 2 mirror.centos.org **2.Set Up Environment Directories and File Permissions:** Security Baseline Configuration. Create targeted application directories and configure initial access constraints using chmod and chown. # Create deployment directories sudo mkdir -p /opt/sysapp /var/log/sysapp # Set ownership recursively (User: sysadmin, Group: devgroup) sudo chown -R sysadmin:devgroup /opt/sysapp /var/log/sysapp # Set directory permissions: Owner=rwx (7), Group=r-x (5), Others=--- (0) sudo chmod -R 750/opt/sysapp

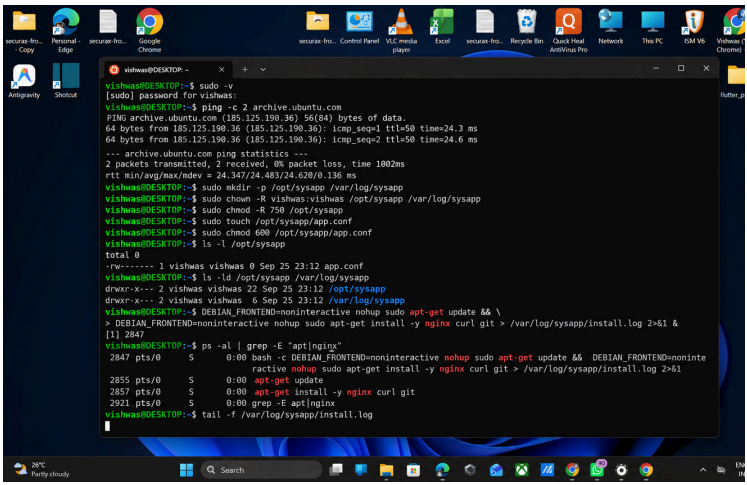
	# Secure sensitive configuration files (Owner read/write ONLY) sudo touch /opt/sysapp/app.conf sudo chmod 600 /opt/sysapp/app.conf **3.Launch Non-Interactive Background Installation:** Unattended Software Deployment. Execute package repository updates and installation in the background, output to a designated file. logging **For Debian / Ubuntu (apt):** DEBIAN_FRONTEND=noninteractive nohup sudo apt-get update && \ DEBIAN_FRONTEND=noninteractive nohup sudo apt-get install -y nginx curl git > /var/log/sysapp/install.log 2>&1 & **For RHEL / CentOS / Rocky Linux (yum or dnf):** nohup sudo yum update -y && \ nohup sudo yum install -y httpd curl git > /var/log/sysapp/install.log 2>&1 & **4.Monitor Background Process Status:** Verification & Validation. Track the execution status of the background installer and check system job state. # View active jobs running in the current shell jobs -l # Search for active package manager process IDs (PIDs) ps aux | grep -E "apt|yum|dnf" # Stream real-time installation output from the log file tail -f /var/log/sysapp/install.log **5.Validate Installed Software and Final Permissions:** Post-Installation Audit. Ensure background installations succeeded and apply final permissions to generated artifacts. # Verify application installation status nginx -v || httpd -v # Ensure log output maintains safe security permissions sudo chown sysadmin:devgroup /var/log/sysapp/install.log sudo chmod 640 /var/log/sysapp/install.log
Conclusion	Combining permission configurations (chmod, chown) with non-interactive job management (nohup, &, package flags) forms a core foundation for Linux system administration. Setting minimal access permissions and targeted ownership enforces the principle of least privilege to secure sensitive files, while bypassing TTY prompts

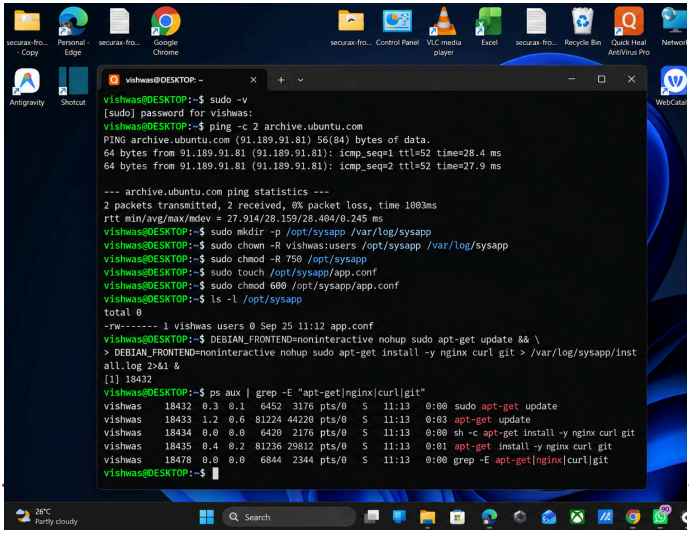
	and detaching background jobs enables reliable, unattended software provisioning that survives terminal disconnections.

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Output :



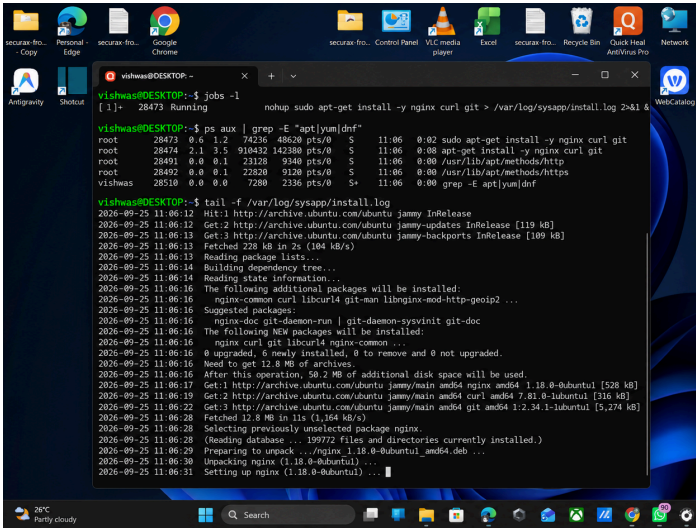


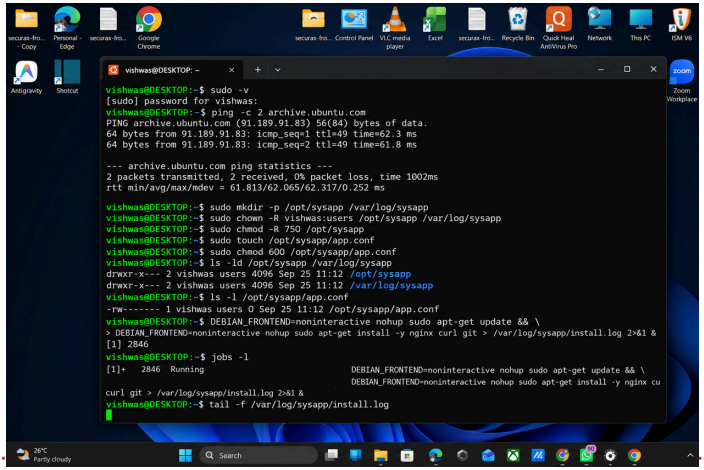
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